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Control apparatus for vehicle

Abstract

A control apparatus for a vehicle that includes (i) a first engagement device configured to connect and disconnect an engine and an electric motor to and from each other, and (ii) a second engagement device configured to connect and disconnect the electric motor and drive wheel to and from each other. In event of a failure of the second engagement device, the control apparatus is configured to forcibly bring the second engagement device into an engaged state and to bring the first engagement device into an engaged state, so as to cause the vehicle to perform a limp-home running in a first running mode in which the vehicle runs using at least the engine. When a running speed of the vehicle becomes lower than a predetermined low-speed threshold value during the limp-home running, the control apparatus is configured to bring the first engagement device into a released state.

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Patent No.	Application Date	Country	CPC
2016151657	12/2015	WO	N/A

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Background/Summary

(1) This application claims priority from Japanese Patent Application No. 2022-175018 filed on Oct. 31, 2022, the disclosure of which is herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

(2) The present invention relates to a control apparatus for a vehicle and, more particularly, to a control executed in event of a failure of a second engagement device disposed between an electric motor and a drive wheel.

BACKGROUND OF THE INVENTION

(3) There is known a hybrid vehicle including a drive wheel, an engine, an electric motor connected to a power transmission path between the engine and the drive wheel in a power transmittable manner, a first engagement device configured to connect and disconnect the engine and the electric motor to and from each other, and a second engagement device configured to connect and disconnect the electric motor and the drive wheel to and from each other, wherein the engine and the electric motor are used as power sources for driving the vehicle. In such a vehicle, it is possible

to switch between a first running mode in which the vehicle runs using at least the engine with both of the first and second engagement devices being in respective engaged states and a second running mode in which the vehicle runs using the electric motor with the first engagement device being in a released state and with the second engagement device being in the engaged state. An example of such a vehicle is disclosed in Patent Document 1. In this Patent Document 1, in event of an ON failure in which the second engagement device (CL2) is maintained in the engaged state, the vehicle is caused to perform a limp-home running in the first running mode (HEV mode) with switching from the first running mode to the second running mode (EV mode) being inhibited, thereby increasing a cruising distance of the limp-home running.

PRIOR ART DOCUMENT

Patent Document

(4) [Patent Document 1] WO2016/151657

SUMMARY OF THE INVENTION

(5) However, in event of the ON failure in which the second engagement device is maintained in the engaged state, if the first engagement device cannot be caused to slip, there is a risk that the engine could stall when a running speed of the vehicle is reduced. In addition, in event of an OFF failure in which the second engagement device is maintained in the released state, power transmission from the power sources (the engine and the electric motor) to the drive wheel is cut off, thereby causing a risk that the vehicle could not run.

(6) The present invention has been made in view of the above circumstances, and an object of the present invention is to enable the vehicle to run in the first running mode while avoiding an engine stall at a low running speed of the vehicle in either case of the ON failure or the OFF failure of the second engagement device.

(7) According to a first aspect of the present invention, there is provided a control apparatus for a vehicle that includes a drive wheel, an engine, an electric motor connected to a power transmission path between the engine and the drive wheel in a power transmittable manner, a first engagement device configured to connect and disconnect the engine and the electric motor to and from each other, and a second engagement device configured to connect and disconnect the electric motor and the drive wheel to and from each other, wherein the engine and the electric motor are used as power sources for driving the vehicle. The control apparatus is configured to switch between a first running mode in which the vehicle runs using at least the engine with both of the first and second engagement devices being in respective engaged states and a second running mode in which the vehicle runs using the electric motor with the first engagement device being in a released state and with the second engagement device being in the engaged state. In event of a failure of the second engagement device, the control apparatus is configured to forcibly bring the second engagement device into the engaged state and to bring the first engagement device into the engaged state, so as to cause the vehicle to perform a limp-home running in the first running mode. When a running speed of the vehicle becomes lower than a predetermined low-speed threshold value during the limp-home running, the control apparatus is configured to bring the first engagement device into the released state.

(8) According to a preferred arrangement of the first aspect of the invention, the second engagement device includes a hydraulically-operated clutch that is to be operated based on a hydraulic pressure supplied to the hydraulically-operated clutch through a first valve that is to be controlled in accordance with a control command signal. The control apparatus is configured to determine whether the second engagement device has failed or not, by comparing a command content of the control command signal and an actual operation state of the hydraulically-operated clutch of the second engagement device. The vehicle includes, in addition to the first valve, a second valve that is to be selectively placed in pressure discharging and charging states, such that the second valve allows the hydraulic pressure to be discharged therethrough from the hydraulically-operated clutch when being placed in the pressure discharging state, and such that the

second valve allows the hydraulic pressure to be charged therethrough to the hydraulically-operated clutch when being placed in the pressure charging state. The control apparatus is configured, when determining that the second engagement device has failed, to forcibly bring the hydraulically-operated clutch of the second engagement device into the engaged state, by placing the second valve in the charging state.

(9) According to a second aspect of the present invention, in the control apparatus according to the first aspect of the present invention, when the running speed of the vehicle is lower than the low-speed threshold value in event of the failure of the second engagement device, the control apparatus is configured to cause the vehicle to perform the limp-home running in the second running mode, by forcibly bringing the second engagement device into the engaged state and bringing the first engagement device into the released state.

(10) In the control apparatus constructed as described above, in event of the failure of the second engagement device, the second engagement device is forcibly brought into the engaged state and the first engagement device is brought into the engaged state to perform the limp-home running in the first running mode in which the engine is used as the power source. Therefore, the cruising distance of the limp-home running can be extended. Further, since the second engagement device is forcibly brought into the engagement state, the limp-home running can be performed in the first running mode not only in event of an ON failure in which the second engagement device is maintained in the engagement state but also in event of an OFF failure in which the second engagement device is maintained in the released state. On the other hand, when the vehicle running speed falls below the predetermined low-speed threshold value during the limp-home running, the first engagement device is brought into the released state. Therefore, the engine is disconnected from the power transmission path, whereby the engine stall is avoided. In this case, for example, it is possible to continue the limp-home running in the second running mode as in the second aspect of the invention, and to extend the cruising distance by resuming the limp-home running in the first running mode when the vehicle running speed becomes equal to or higher than the low-speed threshold value.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic configuration diagram illustrating a vehicle drive system including an electronic control apparatus according to an embodiment of the present invention.

(2) FIG. 2 is a hydraulic circuit diagram illustrating an example of a hydraulic control circuit relating to a WSC clutch included in a vehicle shown in FIG. 1.

(3) FIG. 3 is a flowchart for specifically explaining a control routine executed by a limp-home-running control portion functionally provided in the electronic control apparatus shown in FIG. 1.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENT

(4) The present invention can be applied to various hybrid vehicles each including an engine and an electric motor as power sources and switchable between at least a first running mode in which the vehicle runs using the engine and a second running mode in which the vehicle runs using the electric motor. As the electric motor, a motor generator that also functions as a generator is preferably used, but an electric motor that does not function as a generator can also be used. A hydraulic power transmission apparatus such as a torque converter or an automatic transmission may be provided in a power transmission path between the electric motor and the drive wheels as necessary. When a lock-up clutch (direct coupling clutch) is provided in the hydraulic power transmission apparatus, the lock-up clutch may be used as the second engagement device. A shifting engagement device of an automatic transmission may be used as the second engagement device.

(5) In event of the failure of the second engagement device, the second engagement device is forcibly brought into the engaged state. However, for example, in event of the ON failure in which the second engagement device is maintained in the engaged state, the forcible-engagement control is not necessarily required, and the forcible-engagement control may be performed only in even of the OFF failure in which the second engagement device is maintained in the released state. When the vehicle running speed falls below the predetermined low-speed threshold value, the first engagement device is brought into the released state to avoid the engine stall, and the limp-home running can be continued, for example, in the second running mode. Where a slip engagement control of the first engagement device can be performed, the vehicle can run at a low speed, stop or start in the first running mode by the slip engagement control of the first engagement device.

(6) An embodiment of the present invention will be described in detail below with reference to the drawings.

(7) FIG. 1 is a diagram illustrating a schematic configuration of a drive system of a vehicle **10** including an electronic control apparatus **90** as a control apparatus according to the embodiment of the present invention, and is a diagram illustrating control functions and a main part of a control system for various controls in the vehicle **10**. The vehicle **10** is a hybrid electric vehicle that includes an engine **12** and an electric motor MG as power sources. The vehicle **10** includes the engine **12**, drive wheels **14** that are left and right rear wheels or front wheels, and a power transmission apparatus **18** provided in a power transmission path between the engine **12** and the drive wheels **14**. The engine **12** is an internal combustion engine such as a gasoline engine or a diesel engine. In the engine **12**, an engine control device **22** including a throttle actuator, a fuel injection device and an ignition device is controlled by the electronic control apparatus **90**, so that an engine torque T_e which is a torque of the engine **12** is controlled. The electric motor MG is a rotary electric machine having a function as a motor that generates a mechanical power from an electric power and a function as a generator that generates the electric power from the mechanical power, and is a so-called motor generator. The electric motor MG is connected to a battery **28** via a power control unit (PCU) **24** including an inverter. The PCU **24** is controlled by the electronic control apparatus **90**, whereby an MG torque T_{mg} , which is a torque of the motor MG, is controlled.

(8) The power transmission apparatus **18** includes a K0 clutch **34**, a WSC clutch **36** and an automatic transmission (AT) **38** that are provided in a casing **32** that is a non-rotating member attached to a body of the vehicle **10**, such that the power is transmitted from a transmission output shaft **40** to the drive wheels **14** via a differential gear device **42** and a pair of drive shafts **44**. The K0 clutch **34** is a first engagement device that is configured to connect and disconnect the engine **12** and the motor MG to and from each other, and serves as an engine connection/disconnection clutch that is configured to disconnect the engine **12** from the power transmission path. The WSC clutch **36** as a hydraulically-operated clutch included in a second engagement device that is configured to connect and disconnect the electric motor MG and the drive wheels **14** to and from each other, and serves as a starting clutch or an input clutch by being slip-engagement controlled by the electronic control apparatus **90**.

(9) In the casing **32**, the power transmission apparatus **18** includes an engine connection shaft **46** that connects the engine **12** and the K0 clutch **34** and a motor connection shaft **48** that connects the K0 clutch **34** and the WSC clutch **36**. The motor MG is connected to the motor connection shaft **48** so as to be able to transmit the power. Each of the K0 clutch **34** and the WSC clutch **36** is a wet-type or dry-type frictional engagement device constituted by a multi-disc clutch or a single-disc clutch that is to be pressed by an actuator, and is to be switched between operation states such as an engaged state and a released state by the electronic control apparatus **90**. The operation state of the K0 clutch **34** is switched when a K0 torque T_{k0} , which is a torque capacity of the K0 clutch **34**, is changed by a K0 hydraulic pressure PR_{k0} supplied from a hydraulic control circuit **52**. The operation state of the WSC clutch **36** is switched by when a WSC torque T_{wsc} , which is a torque

capacity of the WSC clutch **36**, is changed by a WSC hydraulic pressure PR_{wsc} supplied from the hydraulic control circuit **52**. An input side member of the WSC clutch **36** is connected to the motor connection shaft **48**, and an output side member of the WSC clutch **36** is connected to a transmission input shaft **50**, which is an input rotary member of the automatic transmission **38**.

(10) The automatic transmission **38** is, for example, a known planetary gear type automatic transmission including a plurality of planetary gear devices and a plurality of engagement devices CB. Each of the engagement devices CB is a hydraulic friction engagement device including, for example, a multi-plate or single-plate clutch or brake that is to be pressed by a hydraulic actuator, a band brake that is to be tightened by a hydraulic actuator, or the like. A CB torque T_{cb} , which is a torque capacity of each of the plurality of engagement devices CB, is changed by the regulated CB hydraulic pressure PR_{cb} supplied from the hydraulic control circuit **52**, whereby an operation state of each of the engagement devices CB is switched between an engaged state and a released state. The automatic transmission **38** is a stepped transmission that establishes one of a plurality of gear positions providing respective different gear ratios γ_{at} (=input rotational speed N_i /output rotational speed N_o) by engaging at least one of the engagement devices CB. The input rotational speed N_i is a rotational speed of the transmission input shaft **50** and is an input rotational speed of the automatic transmission **38**. The input rotational speed N_i is also a rotational speed of the output-side member of the WSC clutch **36**. The output rotational speed N_o is a rotational speed of the transmission output shaft **40**, and is an output rotational speed of the automatic transmission **38**.

(11) The vehicle **10** includes a mechanical hydraulic pump **58** and an electric hydraulic pump **60**. The mechanical hydraulic pump **58** is connected to, for example, the motor connection shaft **48** via a gear, a belt, a chain or the like in a power transmittable manner, and is to be driven by at least one of the engine **12** and the electric motor MG to discharge a working fluid used in the power transmission apparatus **18**. The electric hydraulic pump **60** is to be rotationally driven by a pump motor (not shown) to discharge the working fluid. The working fluid discharged from the mechanical hydraulic pump **58** and the electric hydraulic pump **60** is supplied to the hydraulic control circuit **52**. The hydraulic control circuit **52** outputs the CB hydraulic pressure PR_{cb} , the K0 hydraulic pressure PR_{k0} , the WSC hydraulic pressure PR_{wsc} and the like, which are regulated based on the working fluid discharged from the mechanical hydraulic pumps **58** and the electric hydraulic pumps **60**.

(12) The hydraulic control circuit **52** includes a WSC forcible-engagement portion **54** in relation to an engagement/disengagement control of the WSC clutch **36**. FIG. 2 shows a hydraulic circuit related to the WSC clutch **36** in the hydraulic control circuit **52**, in which the WSC hydraulic pressure PR_{wsc} regulated by a linear solenoid valve SLW as a first valve using a line pressure PL as a source pressure is supplied to the WSC clutch **36** for the engagement/disengagement control. The line pressure PL is a hydraulic pressure serving as the source pressure for various control hydraulic pressures, and is controlled in accordance with, for example, an accelerator opening degree θ_{acc} . The linear solenoid valve SLW is, for example, of a normally open type that is configured, when an excitation current is not supplied to the linear solenoid valve SLW, to output the line pressure PL as it is as the WSC hydraulic pressure PR_{wsc} , so that the WSC clutch **36** is engaged when the excitation current is not supplied to the linear solenoid valve SLW. When the excitation current is supplied to the linear solenoid valve SLW in accordance with a WSC-hydraulic-pressure control command signal S_{wsc} , the WSC hydraulic pressure PR_{wsc} is reduced with increase of a magnitude of the excitation current, so that the WSC torque T_{wsc} is reduced with the increase of the magnitude of the excitation current. When the magnitude of the excitation current becomes equal to or larger than a predetermined value, the WSC clutch **36** is released. A drain passage **55** of the linear solenoid valve SLW is provided with an electromagnetic switch valve **56** as a second valve serving as the WSC forcible-engagement portion **54**.

(13) The electromagnetic switch valve **56** is normally held in a normal connection state as shown in FIG. 2, so as to allow the working fluid is drained from the drain passage **55**. However, when the

exciting current is supplied to the electromagnetic switch valve **56** in accordance with a WSC forcible-engagement command signal S_{comp} outputted from the electronic control apparatus **90**, the electromagnetic switch valve **56** is switched to a forcible-engagement connection state (pressure charging state) from the above-described normal connection state (pressure discharging state), so that the line pressure PL is supplied to the drain passage **55** to forcibly engage the WSC clutch **36**. That is, even if the linear solenoid valve SLW is maintained in the closed state (drain state) and output of the WSC hydraulic pressure PR_{wsc} from the linear solenoid valve SLW is stopped, due to, for example, a short circuit in an electric circuit of the linear solenoid valve SLW , the line pressure PL is supplied from the electromagnetic switch valve **56** to the WSC clutch **36** via the drain passage **55** and the linear solenoid valve SLW by switching the electromagnetic switch valve **56** to a forcible-engagement connection state, whereby the WSC clutch **36** is forcibly engaged. The linear solenoid valve SLW may be also a normally-closed type in which the output of the WSC hydraulic pressure PR_{wsc} is stopped when the excitation current is not supplied to the linear solenoid valve SLW . Although not shown, the $K0$ clutch **34** is also subjected to the engagement/disengagement control by controlling the $K0$ hydraulic pressure PR_{k0} outputted from a linear solenoid valve in accordance with a $K0$ -hydraulic-pressure control command signal Sk_0 , for example, similarly to the WSC clutch **36**.

(14) The vehicle **10** includes the electronic control apparatus **90** as a control apparatus configured to execute various controls. The electronic control apparatus **90** includes a so-called microcomputer including, for example, a CPU, a RAM, a ROM and an input/output interface, and is configured to execute the various controls in the vehicle **10**. The electronic control apparatus **90** includes a plurality of computers for an engine control, an MG control, a hydraulic pressure control and the like as necessary.

(15) The electronic control apparatus **90** is configured to receive various information necessary for the various controls, for example, from an engine speed sensor **70**, an input speed sensor **72**, an output speed sensor **74**, an MG speed sensor **76**, an accelerator-opening degree sensor **80**, a throttle-opening degree sensor **82**, a brake switch **84**, a battery sensor **86** and a fluid temperature sensor **88**, for example, wherein the various information include an engine rotational speed N_e that is a rotational speed of the engine **12**, an input rotational speed N_i , an output rotational speed N_o corresponding to a running speed V of the vehicle **10**, an MG rotational speed N_{mg} that is a rotational speed of the electric motor MG , an accelerator opening degree θ_{acc} which is an amount of operation of an accelerator pedal or the like and which represents an amount of output requested by a vehicle driver of the vehicle **10**, a throttle-valve opening degree θ_{th} that is an opening degree of an electronic throttle valve, a brake ON signal B_{on} that is a signal representing a state in which a brake pedal is being operated by the vehicle driver for operating wheel brakes, a battery temperature TH_{bat} that is a temperature of the battery **28** as an electric storage device, a battery charge/discharge current I_{bat} and a battery voltage V_{bat} of the battery **28**, and an oil temperature TH_{oil} of the working fluid in the hydraulic control circuit **52**. The MG rotational speed N_{mg} is the same as a rotational speed of the input-side member of the WSC clutch **36**, and the input rotational speed N_i is the same as a rotational speed of the output-side member of the WSC clutch **36**. Based on the rotational speeds N_{mg} and N_i , it is possible to determine the operation state of the WSC clutch **36**, that is, whether the WSC clutch **36** is in a fully engaged state, a slip-engaged state or a released state.

(16) The electronic control apparatus **90** outputs an engine control command signal S_e for controlling the engine **12**, an MG control command signal S_{mg} for controlling the electric motor MG , a CB hydraulic-pressure control command signal S_{cb} for controlling the engagement devices CB , a $K0$ hydraulic-pressure control command signal S_{k0} for controlling the $K0$ clutch **34**, a WSC hydraulic-pressure control command signal S_{wsc} for controlling the WSC clutch **36**, a WSC forcible-engagement command signal S_{comp} for forcibly engaging the WSC clutch **36**, and an electric-hydraulic-pump control command signal S_{eop} to various devices provided in the vehicle

10 such as the engine control device **22**, the PCU **24**, the hydraulic control circuit **52** and the electric hydraulic pump **60**. The hydraulic control circuit **52** is provided with the linear solenoid valve SLW that is to be controlled by the WSC hydraulic-control command signal Swsc, the electromagnetic switch valve **56** that is to be switched to the forcible-engagement connection state by the WSC forcible-engagement command signal Scomp, and various solenoid valves for switching fluid passages and controlling hydraulic pressure in accordance with the K0 hydraulic-control command signal Sk0 and the CB hydraulic-control command signal Scb.

(17) The electronic control apparatus **90** functionally includes a power-source control portion **92**, a WSC-engagement control portion **94**, a shift control portion **96**, and a limp-home running control portion **98** in order to realize the various controls in the vehicle **10**.

(18) The power-source control portion **92** calculates a requested drive amount of the vehicle **10** requested by the vehicle driver, for example, by applying the accelerator opening degree θ_{acc} and the vehicle running speed V to a requested drive amount map that represents a pre-stored relationship obtained by experimentation or determined by an appropriate design theory, namely, a predetermined relationship for obtaining the requested drive amount. The requested drive amount is, for example, a requested drive torque $Trdem$ [Nm], a requested drive force $Frdem$ [N], or a requested drive power $Prdem$ [W] of the drive wheels **14**. The power-source control portion **92** calculates a requested input torque $Tidem$ in the transmission input shaft **50** capable of realizing the requested drive amount, by taking account of a transmission loss, an auxiliary load, the gear ratio yat of the automatic transmission **38**, for example, and obtains a target engine torque $Tetgt$ and a target MG torque $Tmtgt$ that make it possible to obtain the requested input torque $Tidem$. Then, the power-source control portion **92** outputs the engine control command signal Se for controlling the engine **12** so as to output the target engine torque $Tetgt$, and also the MG control command signal Smg for controlling the electric motor MG so as to output the target MG torque $Tmtgt$.

(19) For example, when the requested input torque $Tidem$ can be covered only by an output of the electric motor MG, the power-source control portion **92** establishes a BEV (Battery Electric Vehicle) running mode that is a motor running mode in which the vehicle **10** runs by driving the electric motor MG only by the electric power from the battery **28**. In the BEV running mode, the engine **12** is stopped, with the K0 clutch **34** being released and with the WSC clutch **36** being engaged to perform a BEV running using only the motor MG as the power source. In the BEV running mode, the MG torque Tmg is controlled so as to realize the requested input torque $Tidem$. On the other hand, when the requested input torque $Tidem$ cannot be covered without using at least an output of the engine **12**, the power-source control portion **92** establishes an HEV (Hybrid Electric Vehicle) mode that is an engine running mode. In the HEV running mode, both the K0 clutch **34** and the WSC clutch **36** are engaged to perform an engine running using at least the engine **12** as the power source, i.e., an HEV running. In the HEV running mode, the engine torque Te is controlled so as to realize all or a part of the requested input torque $Tidem$, and the MG torque Tmg is controlled so as to compensate for a torque portion insufficient by the engine torque Te with respect to the requested input torque $Tidem$. In the HEV drive mode, for example, the engine **12** may be operated on the optimum fuel-consumption line, and excess or deficiency of the engine torque Te with respect to the requested input torque $Tidem$ may be increased or decreased by the MG torque Tmg , which is the torque of the electric motor MG. The BEV running mode corresponds to a second running mode, and the HEV running mode corresponds to a first running mode.

(20) When the vehicle **10** is in a stopped state, the WSC-engagement control portion **94** executes a control for maintaining the WSC clutch **36** in the released state or slip-engaged state. By bringing the WSC clutch **36** into the slip-engaged state in the stopped state of the vehicle **10**, it is possible to cause the vehicle **10** to perform a so-called creep running that causes a creep phenomenon in which the vehicle **10** is moved slowly, for example, when a brake OFF operation is performed while the accelerator is OFF. While the vehicle **10** is running in the BEV running mode or the HEV running

mode, the WSC clutch **36** is controlled so as to be maintained in the fully engaged state. However, when the input rotational speed N_i is lower than an operable rotational speed of the engine **12** during running of the vehicle **10** in the HEV running mode at a low running speed, the WSC clutch **36** is controlled to be in the slip-engaged state.

(21) The shift control portion **96** determines whether the automatic transmission **38** is to be shifted or not, for example, by using a shift map that is a predetermined relationship, and outputs the CB hydraulic-pressure control command signal Scb for switching the gear position of the automatic transmission **38** as necessary. The shift map is a predetermined relationship having a shift line for determining whether the automatic transmission **38** is to be shifted or not, on a two dimensional coordinate system having the vehicle running speed V and the requested driving torque $Trdem$ as variables, for example.

(22) When the WSC clutch **36** fails, the limp-home-running control portion **98** performs a limp-home running so as to extend a cruising distance of the vehicle **10**. To be more specific, the limp-home-running control portion **98** performs the limp-home running in accordance with a control routine including steps **S1** to **S8**, which is shown in a flowchart of FIG. **3**. In the flowchart of FIG. **3**, “YES” and “NO” in each of determination steps **S1** and **S4** (represented by rhombus shapes) mean affirmative determination and negative determination, respectively.

(23) The control routine is initiated with step **S1** at which the limp-home-running control portion **98** determines whether or not the WSC clutch **36** has failed. When the WSC clutch **36** has not failed, the control flow goes to step **S2** at which the limp-home-running control portion **98** executes a normal running of the vehicle **10**. The failure of the WSC clutch **36** is an ON failure in which the WSC clutch **36** is maintained in the engaged state, or an OFF failure in which the WSC clutch **36** is maintained in the released state, for example, due to disconnection or short circuit in the electric circuit of the linear solenoid valve **SLW**. Therefore, the limp-home-running control portion **98** can determine whether or not the WSC clutch **36** has failed, for example, by comparing the WSC-hydraulic-pressure control command signal $Swsc$ with the operation state of the WSC clutch **36** such as a differential rotation ΔN_{mi} ($=N_{mg}-N_i$) between the MG rotational speed N_{mg} and the input rotational speed N_i . During the normal running at step **S2**, the vehicle **10** runs with the HEV running mode and the BEV running mode being switched, by the power-source control portion **92**, depending on the vehicle running speed V and the requested drive torque $Trdem$, for example.

(24) When it is determined at step **S1** that the WSC clutch **36** has failed, the control flow goes to step **S3** at which the limp-home-running control portion **98** forcibly brings the WSC clutch **36** into the engaged state. Specifically, by outputting the WSC forcible-engagement command signal $Scomp$, the electromagnetic switch valve **56** of the WSC forcible-engagement portion **54** is switched to the forcible-engagement connection state (pressure charging state) from the normal connection state (pressure discharging state). Thus, in event of the OFF-failure in which the normally-open linear solenoid valve **SLW** is closed due to a short circuit or the like in the electric circuit, the WSC hydraulic pressure PR_{wsc} is drained from the drain passage **55** whereby the WSC clutch **36** is maintained in the released state, the line pressure PL is supplied from the electromagnetic switch valve **56** to the WSC clutch **36** via the drain passage **55** and the linear solenoid valve **SLW** so that the WSC clutch **36** is forcibly engaged by the line pressure PL . In event of the ON-failure in which the linear solenoid valve **SLW** is opened due to disconnection or the like, the line pressure PL is directly supplied from the linear solenoid valve **SLW** to the WSC clutch **36** whereby the WSC clutch **36** is maintained in the engaged state by the line pressure PL , the WSC clutch **36** is maintained in the engaged state regardless of whether or not the line pressure PL is supplied from the electromagnetic switch valve **56**. That is, the WSC clutch **36** is brought into the engaged state regardless of the ON failure or the OFF failure.

(25) Step **S3** is followed by step **S4** at which the limp-home-running control portion **98** determines whether or not the vehicle running speed V is equal to or higher than a predetermined low-speed threshold value V_s . When the vehicle running speed V is equal to or higher than the predetermined

low-speed threshold value V_s , the control flow goes to step S5 at which the limp-home-running control portion **98** maintains the K0 clutch **34** in the engaged state. Step S5 is followed by step S6 at which the limp-home-running control portion **98** performs the normal running. The low-speed threshold value V_s is a running speed value at which the K0 clutch **34** is released in the control of the normal running, for example, a low running speed at which the engine **12** is likely to stall. The normal running at step S6 is the limp-home running performed when the WSC clutch **36** fails. Since the K0 clutch **34** is maintained in the engaged state at step S5, the vehicle **10** runs substantially in the HEV running mode that uses the engine **12** as the power source. In the HEV running mode, for example, it is also possible to charge the battery **28** by performing a regenerative control of the electric motor MG as necessary while causing the vehicle **10** to run using the engine **12** as the power source such that the battery **28** is maintained in a fully charged state.

(26) When the vehicle running speed V is reduced and becomes lower than the low-speed threshold value V_s , namely, when a negative determination is made at step S4, the limp-home-running control portion **98** releases the K0 clutch **34** at step S7 and performs the limp-home running of the vehicle **10** in the BEV running mode using the motor MG as the power source at step S8. By releasing the K0 clutch **34** at step S7, K0, the engine **12** is prevented from stalling. Further, since the engine **12** is stopped in the BEV running mode at step S8, fuel consumption is suppressed. When the vehicle running speed V becomes equal to or higher than the low-speed threshold value V_s , for example, upon re-acceleration or start of the vehicle **10** from the stopped state, an affirmative determination is made at step S4, the vehicle **10** is returned to the limp-home running in the HEV running mode using the engine **12** as the power source at steps S5 and S6.

(27) As described above, according to the limp-home-running control portion **98** functionally provided in the electronic control apparatus **90** of the present embodiment, in event of failure of the WSC clutch **36**, the WSC clutch **36** is forcibly engaged and the K0 clutch **34** is engaged to perform the limp-home running in the HEV running mode using the engine **12** as the power source at steps S3, S4, S5 and S6, so that the cruising distance of the limp-home running can be extended. In addition, since the WSC clutch **36** is forcibly brought into the engaged state, it is possible to perform the limp-home running in the HEV running mode not only in event of the ON failure in which the WSC clutch **36** is maintained in the engaged state but also in event of the OFF failure in which the WSC clutch **36** is maintained in the released state.

(28) On the other hand, when the vehicle running speed V becomes lower than the predetermined low-speed threshold value V_s during the limp-home running (the negative determination at step S4), the K0 clutch **34** is brought into the released state at step S7, so that the engine **12** is disconnected from the power transmission path whereby the engine stall is avoided. In this case, the limp-home running can be continued in the BEV running mode at step S8, and when the vehicle running speed V becomes equal to or higher than the low-speed threshold value V_s , the limp-home running in the HEV running mode is resumed at step S6 so as to extend the cruising distance.

(29) Although the embodiment of the present invention has been described in detail with reference to the drawings, the embodiment is merely an embodiment, and the present invention can be carried out with various modifications and improvements based on the knowledge of those skilled in the art.

NOMENCLATURE OF ELEMENTS

(30) **10**: vehicle **12**: engine **14**: drive wheel **18**: power transmission apparatus (power transmission path) **34**: K0 clutch (first engagement device) **36**: WSC clutch (second engagement device, hydraulically-operated clutch) **56**: electromagnetic switch valve (second valve) **90**: electronic control apparatus (control apparatus) **98**: limp-home-running control portion MG: motor SLW: linear solenoid valve (first valve) Swsc: WSC-hydraulic-pressure control command signal (control command signal) V : vehicle running speed V_s : low-speed threshold value

Claims

1. A control apparatus for a vehicle that includes a drive wheel, an engine, an electric motor connected to a power transmission path between the engine and the drive wheel in a power transmittable manner, a first engagement device configured to connect and disconnect the engine and the electric motor to and from each other, and a second engagement device configured to connect and disconnect the electric motor and the drive wheel to and from each other, wherein the engine and the electric motor are used as power sources for driving the vehicle, wherein the control apparatus is configured to switch between a first running mode in which the vehicle runs using at least the engine with both of the first and second engagement devices being in respective engaged states and a second running mode in which the vehicle runs using the electric motor with the first engagement device being in a released state and with the second engagement device being in the engaged state, wherein, in event of a failure of the second engagement device, the control apparatus is configured to forcibly bring the second engagement device into the engaged state and to bring the first engagement device into the engaged state, so as to cause the vehicle to perform a limp-home running in the first running mode, and wherein, when a running speed of the vehicle becomes lower than a predetermined low-speed threshold value during the limp-home running, the control apparatus is configured to bring the first engagement device into the released state.
 2. The control apparatus according to claim 1, wherein, when the running speed of the vehicle is lower than the low-speed threshold value in event of the failure of the second engagement device, the control apparatus is configured to cause the vehicle to perform the limp-home running in the second running mode, by forcibly bringing the second engagement device into the engaged state and bringing the first engagement device into the released state.
 3. The control apparatus according to claim 1, wherein the second engagement device includes a hydraulically-operated clutch that is to be operated based on a hydraulic pressure supplied to the hydraulically-operated clutch through a first valve that is to be controlled in accordance with a control command signal, wherein the control apparatus is configured to determine whether the second engagement device has failed or not, by comparing a command content of the control command signal and an actual operation state of the hydraulically-operated clutch of the second engagement device, wherein the vehicle includes, in addition to the first valve, a second valve that is to be selectively placed in pressure discharging and charging states, such that the second valve allows the hydraulic pressure to be discharged therethrough from the hydraulically-operated clutch when being placed in the pressure discharging state, and such that the second valve allows the hydraulic pressure to be charged therethrough to the hydraulically-operated clutch when being placed in the pressure charging state, and wherein the control apparatus is configured, when determining that the second engagement device has failed, to forcibly bring the hydraulically-operated clutch of the second engagement device into the engaged state, by placing the second valve in the pressure charging state.
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